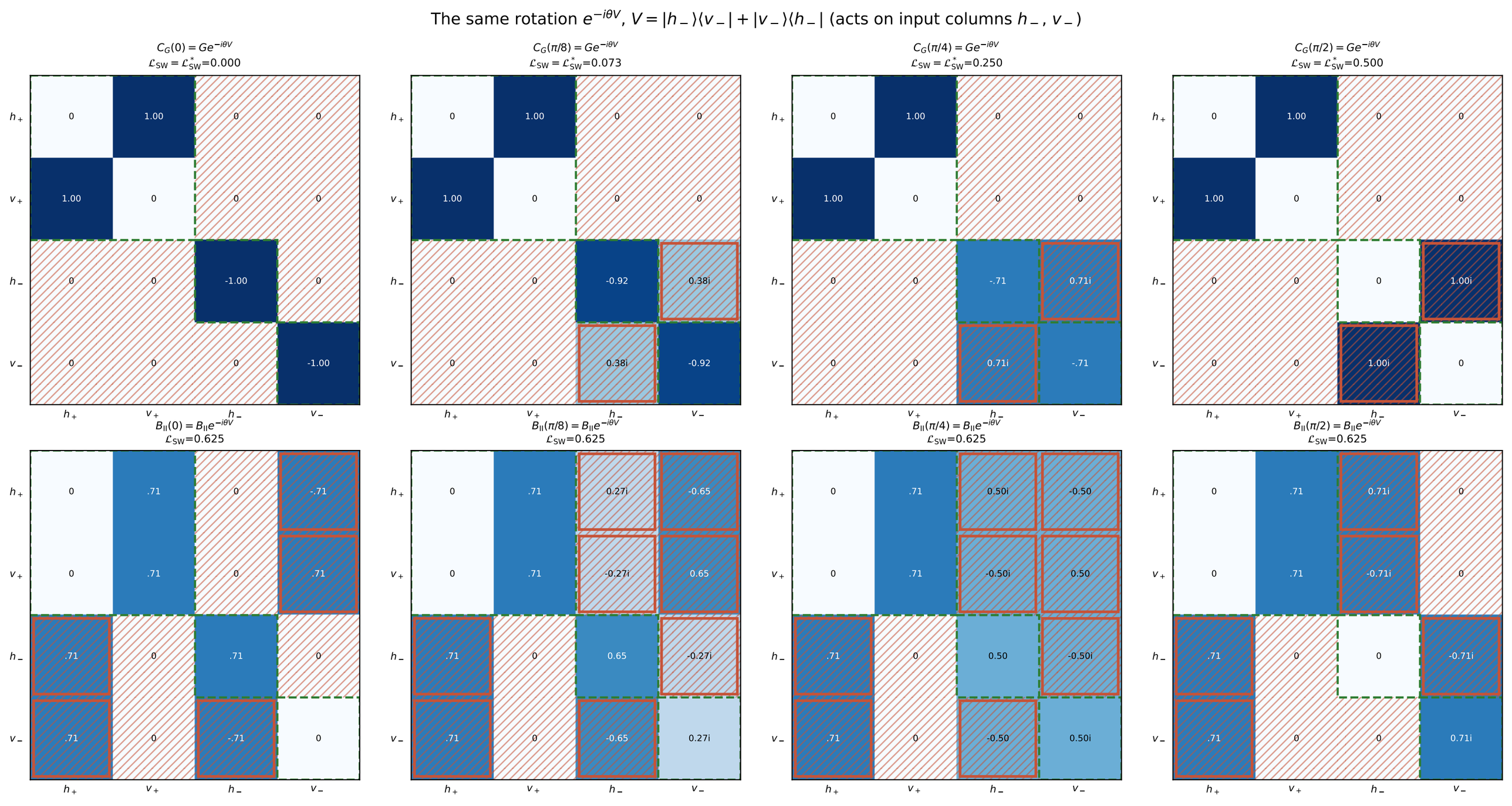
The same rotation $e^{-i\theta V}$, $V = |h_-\rangle\langle v_-| + |v_-\rangle\langle h_-|$ (acts on input columns h_-, v_-)
$C_G(0) = Ge^{-i\theta V}$
$\mathcal{L}_{SW} = \mathcal{L}^*_{SW}$=0.000
$C_G(\pi/8) = Ge^{-i\theta V}$
$\mathcal{L}_{SW} = \mathcal{L}^*_{SW}$=0.073
$C_G(\pi/4) = Ge^{-i\theta V}$
$\mathcal{L}_{SW} = \mathcal{L}^*_{SW}$=0.250
$C_G(\pi/2) = Ge^{-i\theta V}$
$\mathcal{L}_{SW} = \mathcal{L}^*_{SW}$=0.500
$B_{II}(0) = B_{II}e^{-i\theta V}$
$\mathcal{L}_{SW}$=0.625
$B_{II}(\pi/8) = B_{II}e^{-i\theta V}$
$\mathcal{L}_{SW}$=0.625
$B_{II}(\pi/4) = B_{II}e^{-i\theta V}$
$\mathcal{L}_{SW}$=0.625
$B_{II}(\pi/2) = B_{II}e^{-i\theta V}$
$\mathcal{L}_{SW}$=0.625
h_+
v_+
h_-
v_-